1 Planning (15 marks)

Defining the problem (3 marks)

P1 Vary l or l is the independent variable [Allow M] [1]

P2 Determine the period T (for each l [M]) or T is the dependent variable [1]

P3 M is kept constant [l is kept constant] [1]

Methods of data collection (5 marks)

M1 Diagram showing the cantilever is fixed e.g. g-clamp & bench, retort stand & clamp [1]

M2 Many oscillations repeated to determine average T (n $\geqslant$ 10 or $t \geqslant 10$ s for stopwatch) [1]

M3 Weigh M using balance [1]

M4 Measure w and d and measure/record l [1]

M5 Use of vernier caliper/micrometer to measure d and/or w [1]

Method of analysis (2 marks)

A1 Appropriate graph plotted i.e. T^2 against l^3; [T^2 against M] or lg T against lg l or lg M [1]

A2 $E = \dfrac{16\pi^2 M}{wd^3 \times gradient} = \dfrac{k}{gradient}$ $E = \dfrac{16\pi^2 l^3}{wd^3 \times gradient}$

Allow logarithmic solutions e.g. $E = \dfrac{k}{10^{2xy\text{-intercept}}} = \dfrac{k}{100^{y\text{-intercept}}}$ [1]

Safety considerations (1 mark)

S1 Relevant safety precaution related to the use of loads [1]
e.g. cushion/sand in case load falls, keep feet away, keep distance from experiment.

Additional detail (4 marks)

D Relevant points might include [4]

1. Use same rule or keep w and/or d constant.
2. Repeat measurements of d and/or w along rule and average.
3. Discussion of use of motion sensor e.g. orientation or light gates with detail.
4. Use small amplitude or small angle oscillations (to ensure equation is valid).
5. Method of securing load to rule e.g. with tape/glue.
6. Discussion of magnitude of load: large enough to make T large enough.
7. Use of fiducial marker to help to time.
8. Start timing after oscillations have settled.

Do not allow vague use of computers/light gates, video cameras, dataloggers.

[Total: 15]

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